(*b*) Diagrams and representations

Students should be taught, as appropriate, to construct, draw, use and understand:	
the need for correct and precise labelling of all forms of diagrams;	The labelling and scaling of axes will be expected.
pictograms, bar charts, multiple or composite bar charts and pie charts for qualitative, quantitative and discrete data;	The reasons for choosing one form of representation will be expected.
vertical line (stick) graphs for discrete data;	Comparative line graphs will be expected.
for continuous data the following: pie charts, grouped frequency diagrams with equal class intervals, frequency diagrams, cumulative frequency diagrams, population pyramids;	No distinction will be made between cumulative frequency polygons and curves whilst frequency polygons could be open or closed.
stem and leaf diagrams for discrete and continuous data;	Students may need to define the stem for themselves. A key will be expected.
scatter diagrams for bivariate data;	Students may be required to define their own scales.
line graphs and time series;	Trend lines by eye and seasonal variation will be expected.

Content	Notes
choropleth maps (shading);	Eg showing temperature across Europe by shading regions.
simple properties of the shape of distributions of data including symmetry, positive and negative skew;	
the distinction between well presented and poorly presented data;	Poorly presented data can be misleading.
the shape and simple properties of frequency distributions; symmetrical positive and negative skew;	
the potential for visual misuse, by omission or misrepresentation;	Knowing about causes such as unrepresentative scales will be expected.
the transformation from one presentation to another;	Bar chart to pie chart, etc.
how to discover errors in data and recognise data that do not fit a general trend or pattern.	Analytical definition of an outlier will not be required.